STAT 639 - Homework 1

ISLR Chapter 2: Exercises 1, 2, 8, and 10

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Contents

1 Exercise 1

For each case, determine whether a more flexible statistical learning method would generally be expected to perform better or worse than a less flexible method.

1.1 (a) Large number of observations, small number of predictors

A **more flexible method should generally perform better**.

When n is very large and p is small, there is enough data to estimate a flexible model without suffering too much from high variance. A flexible model can capture more complicated relationships between the predictors and response, so the reduction in bias may outweigh the increase in variance.

1.2 (b) Small number of observations, large number of predictors

A **more flexible method should generally perform worse**.

With small n and large p, a flexible method can easily fit random noise in the training data. This leads to overfitting and high variance. A less flexible method is generally safer in this setting.

1.3 (c) Highly non-linear relationship between predictors and response

A **more flexible method should generally perform better**.

If the true relationship is highly non-linear, a flexible model is better able to approximate that relationship. A simple model may have high bias because it cannot capture the true structure.

1.4 (d) Very high variance of the error terms

A **more flexible method should generally perform worse**.

When the irreducible noise is very large, a highly flexible method may begin fitting the noise rather than the underlying signal. This increases variance and can hurt test-set performance.

2 Exercise 2

For each situation, identify whether the problem is classification or regression, whether the main goal is inference or prediction, and identify n and p.

2.1 (a) CEO salary

The response is CEO salary, and the explanatory variables include company profit, number of employees, and industry.

- **Type:** Regression
- **Goal:** Inference
- n**:** 500 firms
- p**:** 3 predictors

This is a regression problem because CEO salary is quantitative. The purpose is to understand which factors are associated with CEO salary, so the primary goal is inference.

2.2 (b) Success of a new product

The response indicates whether a product is successful or unsuccessful. The predictors consist of price, marketing budget, competitors' price, and ten other variables.

- **Type:** Classification
- **Goal:** Prediction
- n**:** 20 products
- p**:** 13 predictors

This is classification because the response is qualitative. The goal is to predict whether the new product will succeed.

2.3 (c) Weekly change in the U.S. dollar

The response is the weekly percentage change in the U.S. dollar relative to the euro, using weekly percentage changes in the stock market and the company's market as predictors.

- **Type:** Regression
- **Goal:** Prediction

- n: 52 weeks
- p: 2 predictors

The response is quantitative, so this is regression. Since the goal is to forecast the weekly percentage change, the main goal is prediction.

3 Exercise 8 - College Data

This exercise uses the `College` data set described in ISLR Chapter 2.

3.1 (a) Read the data

The code below first looks for `College.csv` in the current working directory. If it is not present, it downloads the file from the book website.

```
college_file <- "College.csv"

if (!file.exists(college_file)) {
  download.file(
    "https://www.statlearning.com/s/College.csv",
    destfile = college_file,
    mode = "wb"
  )
}

college <- read.csv("College.csv", check.names = FALSE)
```

There are 777 colleges and universities. Before removing the institution-name column, the imported data contain 19 columns.

3.2 (b) Use college names as row names

```
# Inspect first column
rownames(college) <- college[, 1]
View(college)
```

After removing the institution-name column, the data contain 777 observations and 19 variables. The college names are now stored as row names rather than as a predictor.

3.3 (c-i) Numerical summary

```
summary(college)
```

```
##                          Private              Apps           Accept     
##  Length:777         Length:777         Min.   :   81   Min.   :   72  
##  Class :character   Class :character   1st Qu.:  776   1st Qu.:  604  
##  Mode  :character   Mode  :character   Median : 1558   Median : 1110  
##                                        Mean   : 3002   Mean   : 2019  
##                                        3rd Qu.: 3624   3rd Qu.: 2424  
##                                        Max.   :48094   Max.   :26330  
##      Enroll       Top10perc       Top25perc      F.Undergrad   
##  Min.   :  35   Min.   : 1.00   Min.   :  9.0   Min.   :  139  
##  1st Qu.: 242   1st Qu.:15.00   1st Qu.: 41.0   1st Qu.:  992  
##  Median : 434   Median :23.00   Median : 54.0   Median : 1707  
##  Mean   : 780   Mean   :27.56   Mean   : 55.8   Mean   : 3700  
```

```
##  3rd Qu.: 902   3rd Qu.:35.00   3rd Qu.: 69.0   3rd Qu.: 4005
##  Max.   :6392   Max.   :96.00   Max.   :100.0   Max.   :31643
##   P.Undergrad        Outstate       Room.Board       Books
##  Min.   :    1.0   Min.   : 2340   Min.   :1780   Min.   :  96.0
##  1st Qu.:   95.0   1st Qu.: 7320   1st Qu.:3597   1st Qu.: 470.0
##  Median :  353.0   Median : 9990   Median :4200   Median : 500.0
##  Mean   :  855.3   Mean   :10441   Mean   :4358   Mean   : 549.4
##  3rd Qu.:  967.0   3rd Qu.:12925   3rd Qu.:5050   3rd Qu.: 600.0
##  Max.   :21836.0   Max.   :21700   Max.   :8124   Max.   :2340.0
##     Personal         PhD            Terminal       S.F.Ratio
##  Min.   : 250   Min.   :  8.00   Min.   : 24.0   Min.   : 2.50
##  1st Qu.: 850   1st Qu.: 62.00   1st Qu.: 71.0   1st Qu.:11.50
##  Median :1200   Median : 75.00   Median : 82.0   Median :13.60
##  Mean   :1341   Mean   : 72.66   Mean   : 79.7   Mean   :14.09
##  3rd Qu.:1700   3rd Qu.: 85.00   3rd Qu.: 92.0   3rd Qu.:16.50
##  Max.   :6800   Max.   :103.00   Max.   :100.0   Max.   :39.80
##   perc.alumni        Expend        Grad.Rate
##  Min.   : 0.00   Min.   : 3186   Min.   : 10.00
##  1st Qu.:13.00   1st Qu.: 6751   1st Qu.: 53.00
##  Median :21.00   Median : 8377   Median : 65.00
##  Mean   :22.74   Mean   : 9660   Mean   : 65.46
##  3rd Qu.:31.00   3rd Qu.:10830   3rd Qu.: 78.00
##  Max.   :64.00   Max.   :56233   Max.   :118.00
```

The variables differ substantially in scale. For example, enrollment and application counts vary widely across schools, as do tuition and instructional expenditures. Graduation rates and percentages such as `Top10perc`, `PhD`, and `perc.alumni` are naturally bounded percentage variables.

3.4 (c-ii) Scatterplot matrix of the first ten variables

```r
# check first 10 columns
str(college[, 1:10])
```

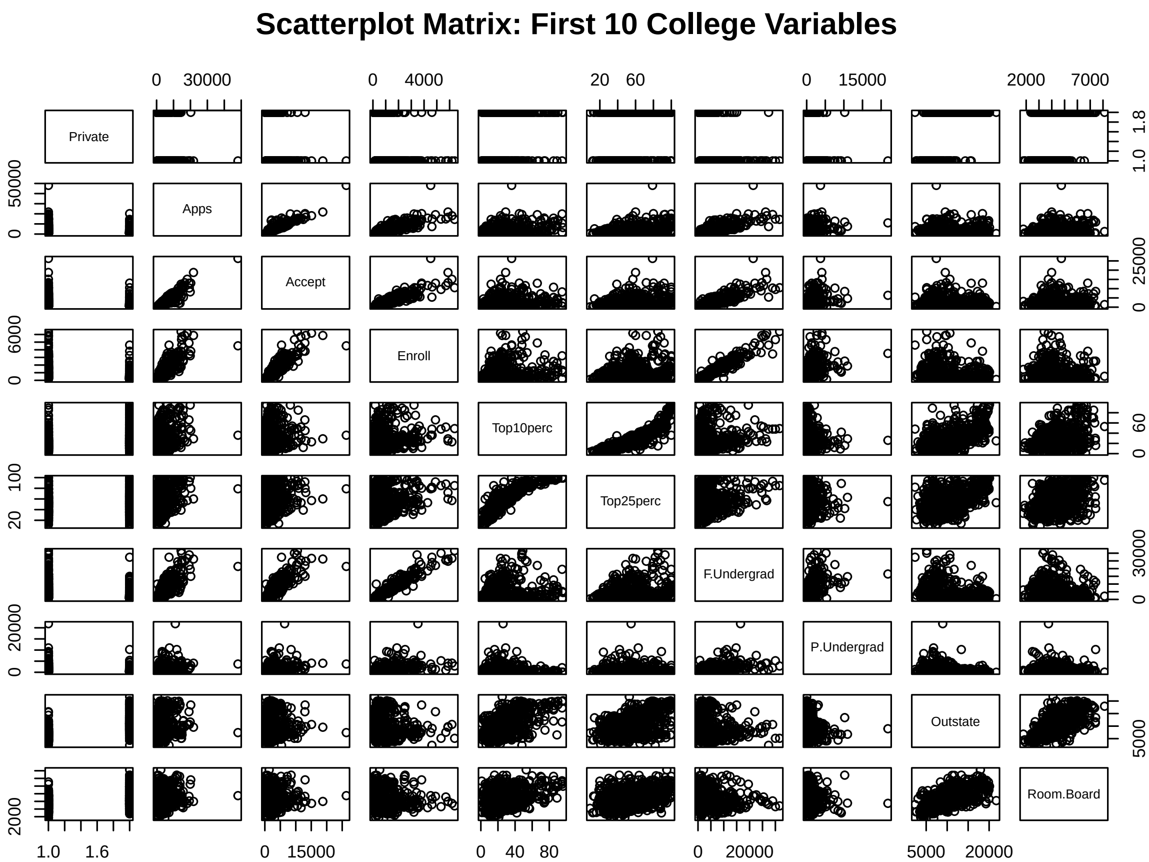
```
## 'data.frame':    777 obs. of  10 variables:
##  $            : chr  "Abilene Christian University" "Adelphi University" "Adrian College" "Agnes Sco
##  $ Private    : chr  "Yes" "Yes" "Yes" "Yes" ...
##  $ Apps       : int  1660 2186 1428 417 193 587 353 1899 1038 582 ...
##  $ Accept     : int  1232 1924 1097 349 146 479 340 1720 839 498 ...
##  $ Enroll     : int  721 512 336 137 55 158 103 489 227 172 ...
##  $ Top10perc  : int  23 16 22 60 16 38 17 37 30 21 ...
##  $ Top25perc  : int  52 29 50 89 44 62 45 68 63 44 ...
##  $ F.Undergrad: int  2885 2683 1036 510 249 678 416 1594 973 799 ...
##  $ P.Undergrad: int  537 1227 99 63 869 41 230 32 306 78 ...
##  $ Outstate   : int  7440 12280 11250 12960 7560 13500 13290 13868 15595 10468 ...
```

```r
rownames(college) <- college[, 1]
college <- college[, -1]

summary(college)
```

```
##    Private               Apps           Accept          Enroll
##  Length:777         Min.   :   81   Min.   :   72   Min.   :  35
##  Class :character   1st Qu.:  776   1st Qu.:  604   1st Qu.: 242
##  Mode  :character   Median : 1558   Median : 1110   Median : 434
##                     Mean   : 3002   Mean   : 2019   Mean   : 780
```

```
##                      3rd Qu.: 3624   3rd Qu.: 2424   3rd Qu.: 902
##                      Max.   :48094   Max.   :26330   Max.   :6392
##    Top10perc       Top25perc      F.Undergrad     P.Undergrad
##  Min.   : 1.00   Min.   :  9.0   Min.   :  139   Min.   :    1.0
##  1st Qu.:15.00   1st Qu.: 41.0   1st Qu.:  992   1st Qu.:   95.0
##  Median :23.00   Median : 54.0   Median : 1707   Median :  353.0
##  Mean   :27.56   Mean   : 55.8   Mean   : 3700   Mean   :  855.3
##  3rd Qu.:35.00   3rd Qu.: 69.0   3rd Qu.: 4005   3rd Qu.:  967.0
##  Max.   :96.00   Max.   :100.0   Max.   :31643   Max.   :21836.0
##     Outstate        Room.Board       Books           Personal
##  Min.   : 2340   Min.   :1780   Min.   :  96.0   Min.   : 250
##  1st Qu.: 7320   1st Qu.:3597   1st Qu.: 470.0   1st Qu.: 850
##  Median : 9990   Median :4200   Median : 500.0   Median :1200
##  Mean   :10441   Mean   :4358   Mean   : 549.4   Mean   :1341
##  3rd Qu.:12925   3rd Qu.:5050   3rd Qu.: 600.0   3rd Qu.:1700
##  Max.   :21700   Max.   :8124   Max.   :2340.0   Max.   :6800
##       PhD            Terminal       S.F.Ratio      perc.alumni
##  Min.   :  8.00   Min.   : 24.0   Min.   : 2.50   Min.   : 0.00
##  1st Qu.: 62.00   1st Qu.: 71.0   1st Qu.:11.50   1st Qu.:13.00
##  Median : 75.00   Median : 82.0   Median :13.60   Median :21.00
##  Mean   : 72.66   Mean   : 79.7   Mean   :14.09   Mean   :22.74
##  3rd Qu.: 85.00   3rd Qu.: 92.0   3rd Qu.:16.50   3rd Qu.:31.00
##  Max.   :103.00   Max.   :100.0   Max.   :39.80   Max.   :64.00
##      Expend         Grad.Rate
##  Min.   : 3186   Min.   : 10.00
##  1st Qu.: 6751   1st Qu.: 53.00
##  Median : 8377   Median : 65.00
##  Mean   : 9660   Mean   : 65.46
##  3rd Qu.:10830   3rd Qu.: 78.00
##  Max.   :56233   Max.   :118.00
```

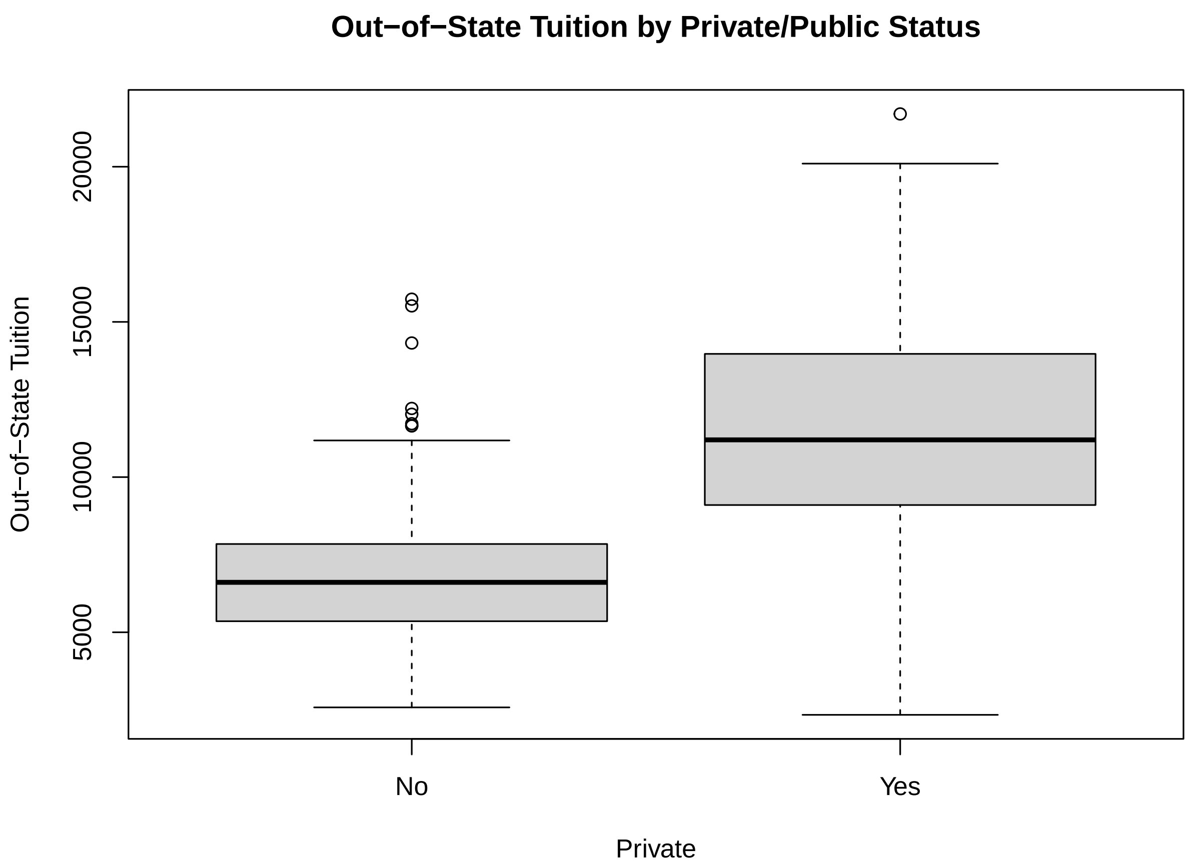
```r
# convert Private to factor before processing
college$Private <- factor(college$Private)

pairs(
  college[, 1:10],
  main = "Scatterplot Matrix: First 10 College Variables"
)
```



Several size-related variables are strongly associated. In particular, colleges receiving more applications also tend to accept and enroll more students. Full-time undergraduate enrollment is also associated with these measures of school size.

3.5 (c-iii) Out-of-state tuition versus private/public status

```r
plot(
  college$Private,
  college$Outstate,
  xlab = "Private",
  ylab = "Out-of-State Tuition",
  main = "Out-of-State Tuition by Private/Public Status"
)
```



Private colleges generally have substantially higher out-of-state tuition than public colleges.

3.6 (c-iv) Create an Elite variable

A school is classified as elite when more than 50% of its students come from the top 10% of their high-school class.

```r
Elite <- rep("No", nrow(college))
Elite[college$Top10perc > 50] <- "Yes"
Elite <- as.factor(Elite)

college <- data.frame(college, Elite)

summary(college$Elite)
```

```
##  No Yes
## 699  78
```

The number of elite colleges is:

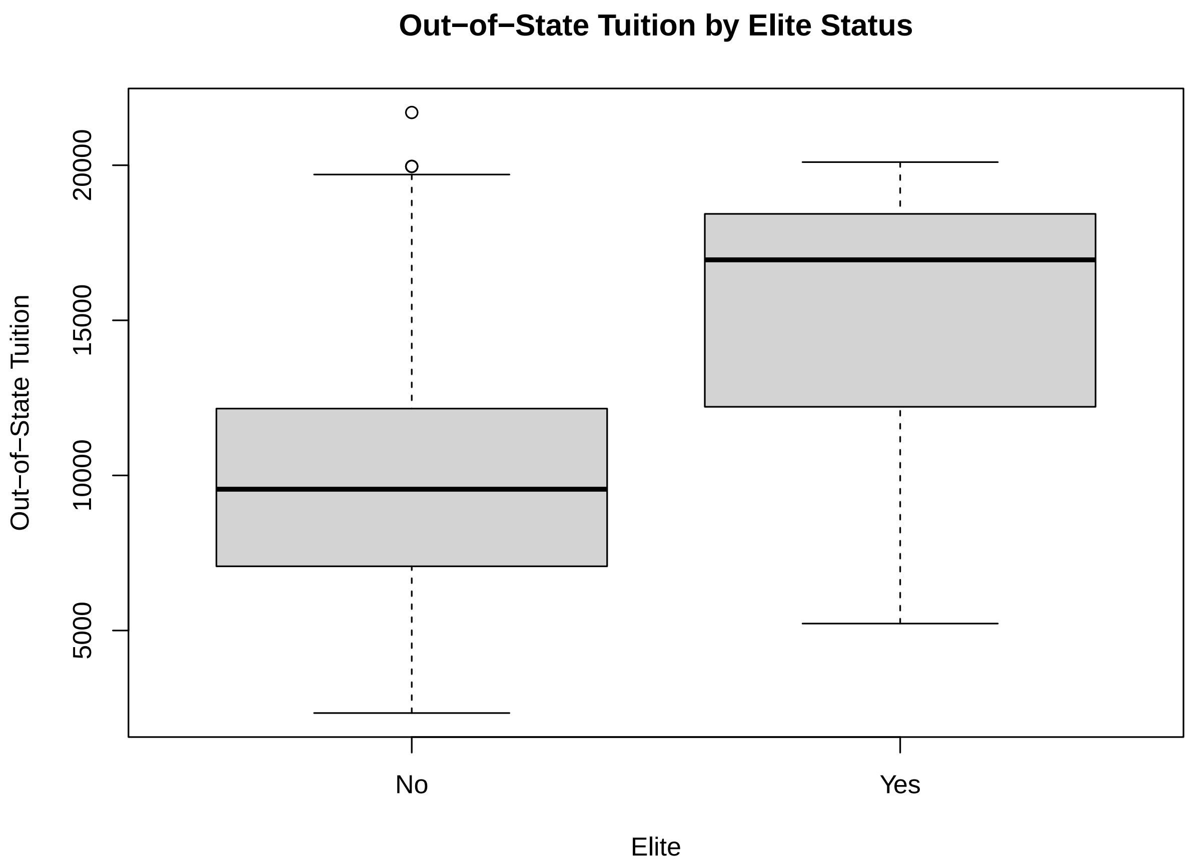
```r
sum(college$Elite == "Yes")
```

```
## [1] 78
```

For the standard ISLR `College` data, this gives **78 elite colleges**.

Now compare tuition between elite and non-elite schools.

```r
plot(
  college$Elite,
  college$Outstate,
  xlab = "Elite",
  ylab = "Out-of-State Tuition",
  main = "Out-of-State Tuition by Elite Status"
)
```

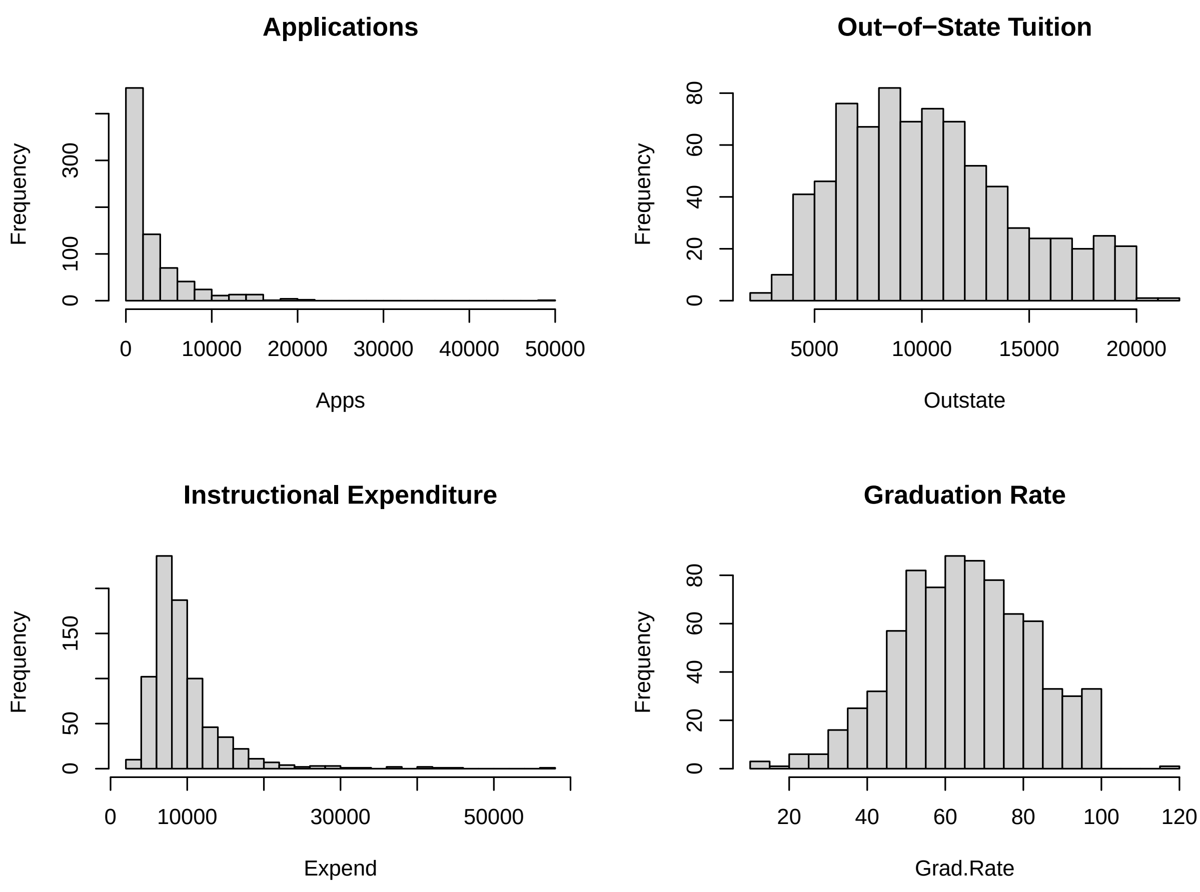


Elite colleges tend to have higher out-of-state tuition than non-elite colleges.

3.7 (c-v) Histograms

```r
par(mfrow = c(2, 2))

hist(
  college$Apps,
  breaks = 20,
  main = "Applications",
  xlab = "Apps"
)

hist(
  college$Outstate,
```

```r
  breaks = 20,
  main = "Out-of-State Tuition",
  xlab = "Outstate"
)

hist(
  college$Expend,
  breaks = 20,
  main = "Instructional Expenditure",
  xlab = "Expend"
)

hist(
  college$Grad.Rate,
  breaks = 20,
  main = "Graduation Rate",
  xlab = "Grad.Rate"
)
```




```r
par(mfrow = c(1, 1))
```

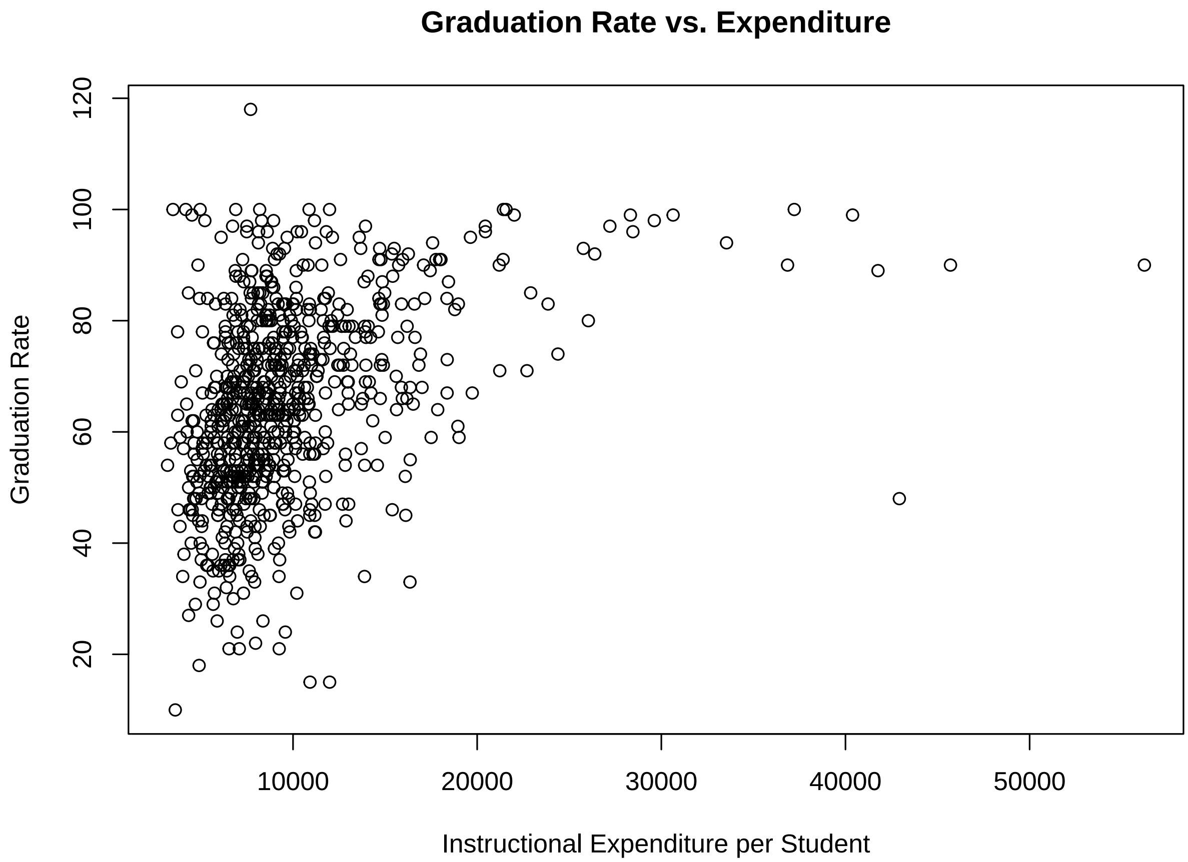
Applications and instructional expenditure show noticeable right-skewness, reflecting a relatively small number of very large or high-spending institutions. Graduation rates are more concentrated toward the upper part of their possible range.

3.8 (c-vi) Additional exploration

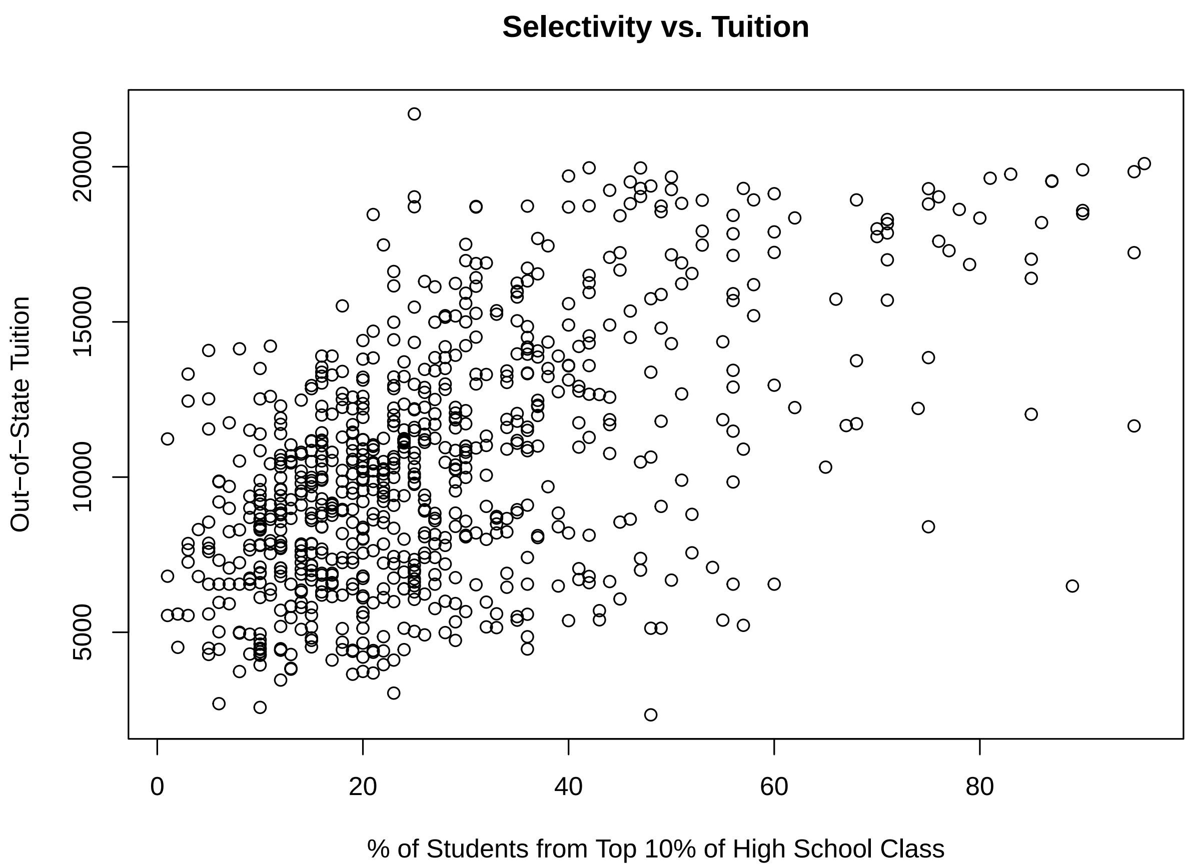
```r
# Correlations among selected quantitative variables
selected <- college[, c(
  "Apps", "Accept", "Enroll", "Top10perc",
  "Outstate", "Room.Board", "PhD",
  "Expend", "Grad.Rate"
)]

round(cor(selected), 2)
```

```
##            Apps Accept Enroll Top10perc Outstate Room.Board  PhD Expend
## Apps       1.00   0.94   0.85      0.34     0.05       0.16 0.39   0.26
## Accept     0.94   1.00   0.91      0.19    -0.03       0.09 0.36   0.12
## Enroll     0.85   0.91   1.00      0.18    -0.16      -0.04 0.33   0.06
## Top10perc  0.34   0.19   0.18      1.00     0.56       0.37 0.53   0.66
## Outstate   0.05  -0.03  -0.16      0.56     1.00       0.65 0.38   0.67
## Room.Board 0.16   0.09  -0.04      0.37     0.65       1.00 0.33   0.50
## PhD        0.39   0.36   0.33      0.53     0.38       0.33 1.00   0.43
## Expend     0.26   0.12   0.06      0.66     0.67       0.50 0.43   1.00
## Grad.Rate  0.15   0.07  -0.02      0.49     0.57       0.42 0.31   0.39
##            Grad.Rate
## Apps            0.15
## Accept          0.07
## Enroll         -0.02
## Top10perc       0.49
## Outstate        0.57
## Room.Board      0.42
## PhD             0.31
## Expend          0.39
## Grad.Rate       1.00
```

```r
# Graduation rate versus instructional expenditure
plot(
  college$Expend,
  college$Grad.Rate,
  xlab = "Instructional Expenditure per Student",
  ylab = "Graduation Rate",
  main = "Graduation Rate vs. Expenditure"
)
```




```r
# Top 10% students versus tuition
plot(
  college$Top10perc,
  college$Outstate,
  xlab = "% of Students from Top 10% of High School Class",
  ylab = "Out-of-State Tuition",
  main = "Selectivity vs. Tuition"
)
```



Overall, the data suggest strong relationships among measures of institution size. More selective colleges also tend to charge higher tuition, and private or elite colleges generally have higher out-of-state tuition.

4 Exercise 10 - Boston Housing Data

4.1 Setup

```
if (!requireNamespace("ISLR2", quietly = TRUE)) {
  install.packages("ISLR2")
}

library(ISLR2)
data("Boston")
```

4.2 (a) Dimensions and meaning of rows and columns

```
?Boston

dim(Boston)
```

```
## [1] 506  13
```

```r
names(Boston)
```

```
##  [1] "crim"    "zn"      "indus"   "chas"    "nox"     "rm"      "age"
##  [8] "dis"     "rad"     "tax"     "ptratio" "lstat"   "medv"
```

```r
str(Boston)
```

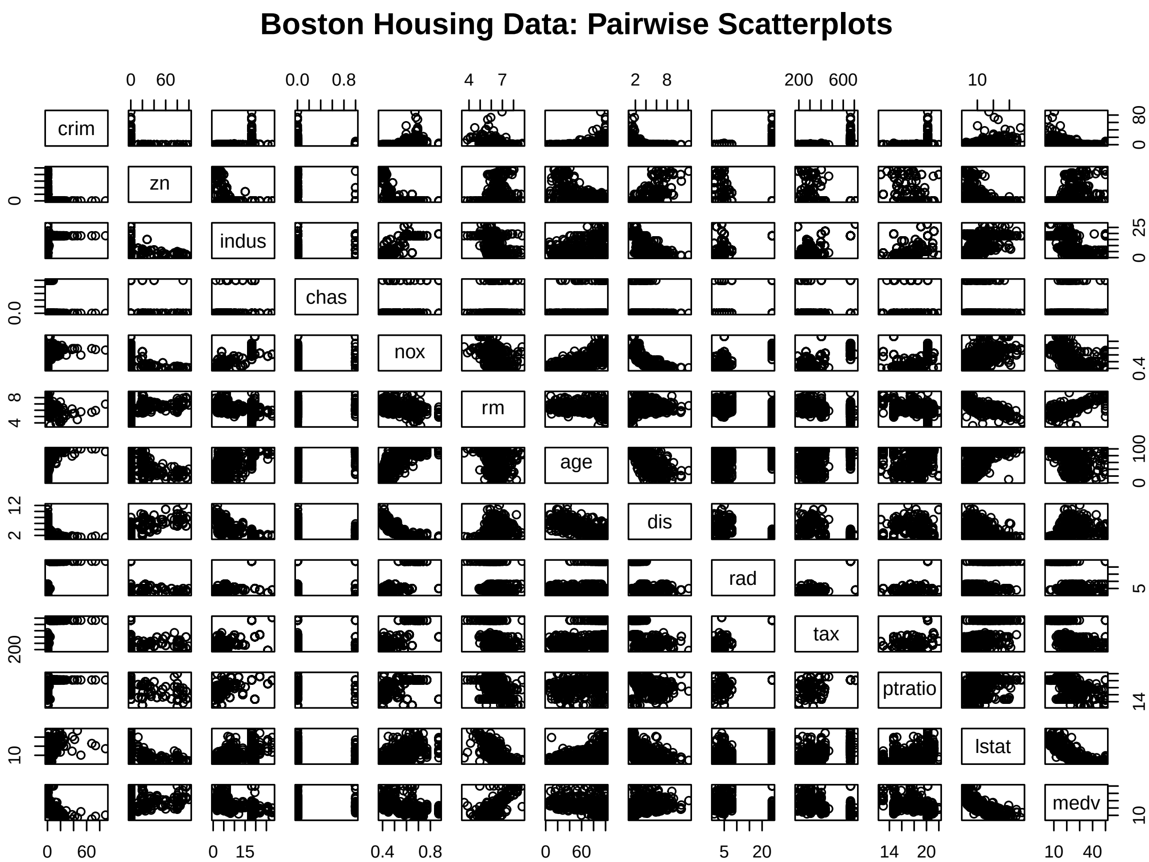
```
## 'data.frame':    506 obs. of  13 variables:
##  $ crim   : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
##  $ zn     : num  18 0 0 0 0 0 12.5 12.5 12.5 12.5 ...
##  $ indus  : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 7.87 ...
##  $ chas   : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ nox    : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 0.524 ...
##  $ rm     : num  6.58 6.42 7.18 7 7.15 ...
##  $ age    : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
##  $ dis    : num  4.09 4.97 4.97 6.06 6.06 ...
##  $ rad    : int  1 2 2 3 3 3 5 5 5 5 ...
##  $ tax    : num  296 242 242 222 222 222 311 311 311 311 ...
##  $ ptratio: num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
##  $ lstat  : num  4.98 9.14 4.03 2.94 5.33 ...
##  $ medv   : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

The Boston data contain **506 rows and 13 columns**.

Each row represents a Boston-area census tract. The columns contain tract-level predictors such as crime rate, zoning, industrial land use, air pollution, average number of rooms, property tax rate, pupil-teacher ratio, and related variables. The `medv` variable is the median value of owner-occupied homes.

4.3 (b) Pairwise scatterplots

```r
pairs(
  Boston,
  main = "Boston Housing Data: Pairwise Scatterplots"
)
```



Some clear relationships are visible. For example, `medv` tends to increase with `rm` and decrease with `lstat`. Crime tends to be higher in areas with higher `rad` and `tax`, and lower in areas with larger `dis`.

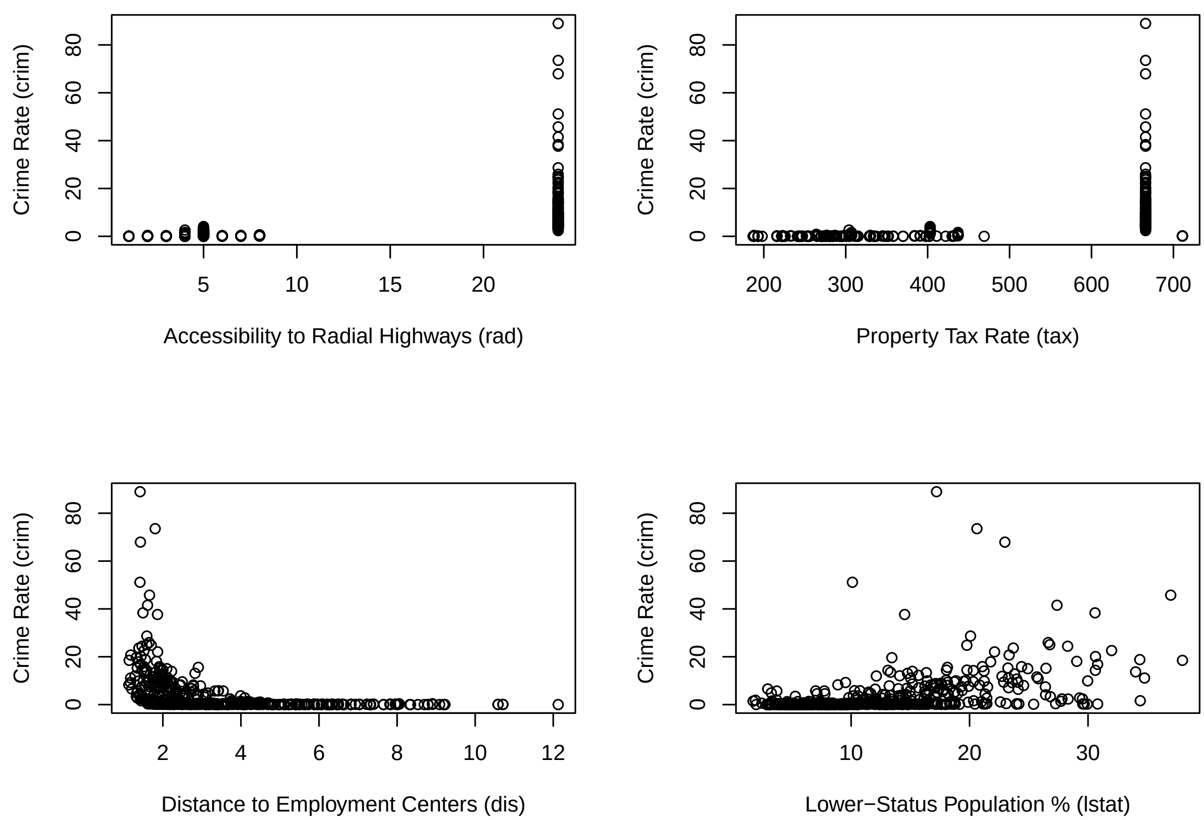
4.4 (c) Predictors associated with per-capita crime rate

```r
crime_cor <- sort(
  cor(Boston)[, "crim"],
  decreasing = TRUE
)

crime_cor
```

```
##        crim         rad         tax       lstat         nox       indus
##  1.00000000  0.62550515  0.58276431  0.45562148  0.42097171  0.40658341
##         age     ptratio        chas          zn          rm         dis
##  0.35273425  0.28994558 -0.05589158 -0.20046922 -0.21924670 -0.37967009
##        medv
## -0.38830461
```

```r
par(mfrow = c(2, 2))

plot(Boston$rad, Boston$crim,
     xlab = "Accessibility to Radial Highways (rad)",
     ylab = "Crime Rate (crim)")
```

```r
plot(Boston$tax, Boston$crim,
     xlab = "Property Tax Rate (tax)",
     ylab = "Crime Rate (crim)")

plot(Boston$dis, Boston$crim,
     xlab = "Distance to Employment Centers (dis)",
     ylab = "Crime Rate (crim)")

plot(Boston$lstat, Boston$crim,
     xlab = "Lower-Status Population % (lstat)",
     ylab = "Crime Rate (crim)")
```




```r
par(mfrow = c(1, 1))
```

Crime rate is positively associated with variables such as `rad`, `tax`, `indus`, `nox`, and `lstat`, and negatively associated with variables such as `dis`, `rm`, and `medv`. These are associations only and should not be interpreted as causal effects.